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# Jordan Barber

Data Scientist

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## Summary

An astrophysics PhD researcher with expertise in computational simulations, statistical modelling, and large-scale data analysis. Looking to transition into a data science role to apply my analytical and programming skills in solving complex, real-world problems. My experience includes a comprehensive understanding of numerical integration techniques, regularisation schemes and simulations including N-Body algorithms, parameterised modelling and Markov Chain Monte Carlo methods.

## Technical Skills

### Software and Programming Skills

***Proficient:***

**Python** : 7 years of experience performing detailed analysis of large simulation data and effectively communicating the results through data visualisation libraries. Developing & running sophisticated simulation codes. Training and deploying a normalising flows ML model for computing posterior distributions of star cluster initial properties. Proficient with scientific/ML libraries, such as NumPy, SciPy, Pandas, Matplotlib and Seaborn, sklearn.
**LaTex** : Writing and formatting academic papers.

***Experienced:***

**Git :** Maintaining and managing both lone and collaborative research projects and simulation codes both locally and remotely on GitHub and Gitlab.
**Bash :** Creating scripts on HPC systems to improve workflow efficiency.
**Unix :** Operate with both git bash and Ubuntu on local machines and through terminals on remote HPC machines.
**High Performance Computing (HPC) :** Configuring and deploying sophisticated simulation codes on supercomputing clusters, Exploiting both CPU parallelisation with MPI and OpenMP processes as well as GPU parallel computing with CUDA.
**SQL :** Querying large distributed datasets through queries and PySpark.
**Azure :** Working with various features of the Azure ecosystem, such as Azure Datalake containers and Databricks for code development.

***Familiar:***

**FORTRAN/c++ :** Implementing a new model prescription in a sophisticated N-Body code to simulate pulsar evolution.

### Useful Skills and Applicable Knowledge

**Statistics :** Comprehensive understanding of high level statistical concepts including but not limited to: regression analysis, hypothesis testing, Bayesian statistics and non-parametric tests.
**Bayesian Statistics :** Detailed understanding of Bayesian statistics and inference used when running posterior estimation via Monte-Carlo Markov Chain sampling for time-series gravitational waves data.
**N-Body Simulations :** Configuring and analysing data from direct N-Body simulation codes from few-body problems up to 2 million particle simulations.
**Monte-Carlo methods :** Utilised in several cluster simulation codes and sampling methods.
**Statistical Modelling :** Produced detailed statistical models of binary black hole properties from different simulation techniques and theoretical evolutionary pathways.
**Machine learning/Normalising flows :** Trained and deployed a machine learning model to compute posterior distributions for initial cluster parameters.
**Technical writing :** Produced 3 major academic papers in scientific journals.
**General presentation skills :** Presented scientific talks at academic conferences, and accessible talks to the public.

## Soft Skills

**Collaboration :** Collaborated with interdisciplinary teams, fostering cooperation to achieve project goals.
**Communication :** Presented technical research at scientific conferences and outreach events, effectively conveying complex ideas to diverse audiences.
**Time Management :** Delivered multiple projects, including publications, teaching, and tool development, on time and to a high standard.
**Leadership :** Supervised student dissertation projects and intern research.
**Critical Thinking & Problem Solving :** Improved and optimized complex simulation codes, ensuring efficiency and accuracy. Developed and implemented new computational methods, such as the C++ pulsar evolution prescription, showcasing my ability to solve technical challenges. Applied Bayesian inference and MCMC methods to tackle high-dimensional parameter estimation problems.

## Experience

**Data Scientist** **Apr 2025 — present**
*Fifty5Blue (Formerly Kantar Media)*

- Developed and maintained predictive pipelines to measure advertiser campaign reach across demographics and digital streaming/social media platforms. This provides crucial insights to our clients, allowing for more strategic ad spending and targeting.
- Led the integration of a major streaming site into an existing cross-media audience measurement product, liaising directly with engineering and product teams to take the solution from idea to production.
- Designed and productionised pipelines in Databricks for computing cross-platform reach and supporting campaign-level reporting across TV and multiple digital platforms.
- Aided in the continued development of a ML modelling pipeline for producing Virtual ID models to our clients. Improving workflows and understanding of training algorithms.
- Integrated two new digital platforms into the VID modelling pipeline, leveraging PySpark and ensuring professional development practices adhered to.

**PhD Researcher – School of Physics and Astronomy** **Sept 2021 — Apr 2025**
*Cardiff University* *Cardiff, UK*

***Star cluster evolution study***

- Implemented and improved upon an existing sophisticated, hybrid N-Body code, written in C++ and FORTRAN and exploiting both CPU and GPU parallisation. Configured the simulation code to run on a supercomputing cluster, producing 34 detailed simulations comprising $\approx$ 10 TB of data. Utilised the Slurm Workload Manager to schedule and manage the simulation runs. Developed and implemented a new C++ stellar evolution prescription to model pulsar evolution.
- Utilised an efficient analytical simulation code to produce $>$ 10 million star cluster simulations. Assisted in the translation of the code from FORTRAN to Python.

***Black hole formation study***

- Deployed parametrised stellar evolution codes to perform statistical modelling of the evolution and possible end states of a population of binary stars. Produced $\approx 100,000$ simulations using code written in Python.

***Normalising flows for star cluster predictions***

- Developing a machine learning model using normalizing flows to compute posterior distributions. The training data is generated from tens of millions of simulations.

***Extra work***

- Monitored and ran real-time Bayesian inference parameter estimation for time-series data. This involves performing a MCMC posterior sampling search over a 15 dimensional parameter space.
- Performed extensive literature reviews both on the astrophysical field and on the computational methods used in modern simulation codes.
- Taught the *Structured Programming*, *Gravitational Wave Astrophysics* and *General Relativity* courses.

**Research Scientist – Modelling observed supernovae** **Aug 2021 — Oct 2021**
*Cardiff University* *Cardiff, UK*

- Developed an open source python tool using Gaussian processes and MCMC sampling to find optimal parameters for modelling supernovae observations.

**Science Communicator** **Sept 2018 — Sept 2020**
*Techniquest* *Cardiff, UK*

- Designed and ran science shows and presentations aimed at the general public.

## Education

**Doctor of Philosophy (PhD) in Physics & Astronomy**, *Cardiff University* March 2025
**Master of Physics (MPhys) in Astrophysics: 1st Class Honours (84%)**, *Cardiff University* July 2021

## Publications

Barber, J., Chattopadhyay, D. and Antonini, F., 2024. Black hole binary mergers in dense star clusters: the importance of primordial binaries. *Monthly Notices of the Royal Astronomical Society*, 527(3), pp.7363-7381.

Barber, J. and Antonini, F., 2025. Formation and evolution of binary black holes in $N$-body simulations of star clusters with up to two million stars. *Monthly Notices of the Royal Astronomical Society*, 538(2), pp.639–658.

Fantoccoli, F., Barber, J., Dosopoulou, F., Chattopadhyay, D. and Antonini, F., 2025. Properties of black hole-star binaries formed in $N$-body simulations of massive star clusters: implications for Gaia black holes. *Monthly Notices of the Royal Astronomical Society*, 538(1), pp.243–257.

Barber, J., 2025. Dynamics of compact objects in dense stellar clusters and the implications for gravitational wave detections. PhD Thesis, Cardiff University.